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Thermodynamic assessments of the Ag–Tb and Ag–Ho systems

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ABSTRACT

The phase diagrams and thermodynamic properties in the Ag–Tb and Ag–Ho binary systems have been assessed by using the CALPHAD (Calculation of Phase Diagrams) method on the basis of the experimental data including the thermodynamic properties and phase equilibria. The Gibbs free energies of the liquid, bcc, fcc and hcp phases were described by the substitution solution model with excess energy expressed by Redlich–Kister polynomial, and those of intermetallic compounds (Ag_2Tb, AgTb, Ag_2Ho and AgHo phases) were treated as stoichiometric compounds, and $Ag_{51}Tb_{14}$ and $Ag_{51}Ho_{14}$ phases were modeled by the sublattice model in these two binary systems. The thermodynamic parameters of each phase in the Ag–Tb and Ag–Ho binary systems were obtained, and an agreement between the calculated results and experimental data was obtained in each binary system.

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1. Introduction

Magnesium (Mg) alloys have been widely used in many industrial fields due to their superior characteristics such as low density, high specific strength and good castability [1]. Addition of rare earths (RE) in Mg base alloys can enhance the high temperature properties, corrosion resistance and casting characteristics [2,3]. Besides, silver addition has a positive effect on properties of Mg base alloys [4]. Therefore, it is important to understand the phase equilibria in the Mg–Ag–RE alloy systems. In the present paper, as a part of the thermodynamic database of the Mg–Ag–RE alloy systems, the thermodynamic descriptions for the phase diagrams in the Ag–Tb and Ag–Ho binary systems were carried out by means of the CALPHAD (Calculation of Phase Diagrams) method, where the thermodynamic parameters to describe the Gibbs free energy of the individual phases are optimized based on the experimental data on thermodynamic properties and phase diagram.

2. Experimental information

2.1. The Ag–Tb system

The Ag–Tb system is consisted of three solid solution phases (fcc (Ag), αTb and βTb phases) and three intermetallic compounds ($Ag_{51}Tb_{14}$, Ag_2Tb and AgTb phases). Mcmasters et al. [5] identified the Ag-rich intermetallic compound to be the $Ag_{51}Tb_{14}$ phase and determined its crystal structure to be the $Ag_{51}Gd_{14}$-type by X-ray powder diffraction methods. Steeb et al. [6] reported the lattice parameters of the $Ag_{51}Ho_{14}$ phase and indicated a small solubility range for this compound. Based on the X-ray parametric method, Gschneidner et al. [7] determined the temperature of the eutectic reaction of the $L \leftrightarrow (Ag) + Ag_{51}Tb_{14}$ ($801 \pm 3\,^{\circ}C$) and the maximum solid solubility of Tb in the fcc (Ag) solution phase (1.12 at.%).

Later, based on metallography, X-ray diffraction, chemical analysis and differential thermal analysis, Delfino et al. [8] investigated the phase diagram in the Ag–Tb system, and reported three stoichiometry compounds of the $Ag_{51}Tb_{14}$ (melting point, 985 °C), Ag_2Tb (melting point, 915 °C) and AgTb (melting point, 1145 °C), and also determined four eutectic reactions: $L \leftrightarrow (Ag) + Ag_{51}Tb_{14}$ at 805 °C, $L \leftrightarrow Ag_{51}Tb_{14} + Ag_2Tb$ at 885 °C, $L \leftrightarrow Ag_2Tb + AgTb$ at 900 °C, and $L \leftrightarrow AgTb + \alpha Tb$ at 860 °C. The part of the phase diagram in the compositional range more than 80 at.% Tb was not determined by Delfino et al. [8].

The solid solubility and the eutectic temperature reported by Gschneidner et al. [7] were accepted in the Ag–Tb binary phase diagram compiled by Gschneidner and Calderwood [9] with some modifications from the experimental data [8].

In addition, Meschel and Kleppa [10] measured the standard enthalpies of formation of the $Ag_{51}Tb_{14}$, Ag_2Tb and AgTb phases by high-temperature direct synthesis calorimetry. Ivanov and Lukashenko [11] measured the standard enthalpy of formation of the AgTb phase, the activities of components and Gibbs free energies of formation of liquid phase at 1090 °C by the Knudsen effusion method. Boer and Perrifor [12] predicted that the enthalpies of mixing of liquid phase in the Ag–Tb system is −29 kJ/mol, however, no temperature dependence was given in the work of Boer and Perrifor [12].

2.2. The Ag–Ho system

The Ag–Ho system is consisted of two solid solution phases (fcc (Ag) and hcp (Ho) phases) and three intermetallic compounds ($Ag_{51}Ho_{14}$, Ag_2Ho and AgHo phases). Wunderlin et al. [13] reported the Ag-rich phase to be the Ag_7Ho_2 and determined the maximum solid solubility of Ho in the fcc (Ag) (1.6 at.%) by using metallographic methods. Wunderlin et al. [14] also determined the eutectic temperature of the $L \leftrightarrow (Ag) + Ag_7Ho_2$ (795 °C).

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Table 1
The stable solid phases and the used models in the Ag–Tb [9] and Ag–Ho [16] systems.

System	Phase	Pearson symbol	Strukturbericht designation	Space group	Prototype
Ag–Tb	(Ag)	cF4	A1	*Fm3m*	Cu
	$Ag_{51}Tb_{14}$	hP65	–	*P6/m*	$Ag_{51}Gd_{14}$
	Ag_2Tb	tI6	C11	*I4/mmm*	$MoSi_2$
	AgTb	cP2	B2	*Pm3m*	CsCl
	(αTb)	hP2	A3	$P6_3/mmc$	Mg
	(βTb)	cI2	A2	*Im3m*	W
Ag–Ho	(Ag)	cF4	A1	*Fm3m*	Cu
	$Ag_{51}Ho_{14}$	hP65	–	*P6/m*	$Ag_{51}Gd_{14}$
	Ag_2Ho	tI6	C11	*I4/mmm*	$MoSi_2$
	AgHo	cP2	B2	*Pm3m*	CsCl
	(Ho)	hP2	A3	$P6_3/mmc$	Mg

Later, based on thermal analysis, metallography and X-ray methods, Gebhardt and Erdberg [15] investigated the phase diagram in the Ag–Ho system, and reported three stoichiometry compounds of the Ag_7Ho_2 (melting point, 920 °C), Ag_2Ho (melting point, 990 °C) and AgHo (melting point, 1165 °C), and also determined the maximum solid solubility of Ho in the fcc (Ag) (1.6 at.% Ho) and four eutectic reactions: L ↔ (Ag) + Ag_7Ho_2 at 775 °C, L ↔ Ag_7Ho_2 + Ag_2Ho at 890 °C, L ↔ Ag_2Ho + AgHo at 970 °C, and L ↔ AgHo + (Ho) at 880 °C.

By using X-ray powder diffraction methods, Mcmasters et al. [5] determined that the correct stoichiometry for the Ag-rich compound with the hexagonal $Ag_{51}Gd_{14}$-type structure is the $Ag_{51}Ho_{14}$ compound, rather than the Ag_7Ho_2 compound. Steeb et al. [6] reported the lattice parameters of the $Ag_{51}Ho_{14}$ compound with a small solubility range. Therefore, the $Ag_{51}Ho_{14}$ phase is accepted in the present work. The eutectic temperature (L ↔ (Ag) + $Ag_{51}Ho_{14}$ at 795 °C) reported by Wunderlin et al. [14] were accepted in the Ag–Ho binary phase diagram compiled by Gschneidner and Calderwood [16] with some modifications from the experimental data [15].

In addition, Fitzner and Kleppa [17] measured the standard enthalpies of formation of the AgHo phase by high-temperature direct synthesis calorimetry. With the same method, Meschel and Kleppa [10] measured the standard enthalpy of formation of the $Ag_{51}Ho_{14}$ and Ag_2Ho phases. Niessen et al. [18] predicted that the standard enthalpies of formation of the $Ag_{51}Ho_{14}$, Ag_2Ho and AgHo phases. Boer and Perrifor [12] predicted that the enthalpies of mixing of liquid phase in the Ag–Ho system is −29 kJ/mol, however, no temperature dependence was given in the work of Boer and Perrifor [12].

3. Thermodynamic models

The information of stable solid phases and the used models in the Ag–Tb and Ag–Ho binary systems is listed in Table 1.

3.1. Solution phases

In the Ag–Me (Me: Tb or Ho) binary systems, Gibbs free energies of the liquid, fcc, bcc and hcp phases are described by the substitution solution model with excess energy expressed by Redlich–Kister polynomial [19], as follows:

$$G_m^{\phi} = \sum_{i=\mathrm{Ag,Me}} {}^0G_i^{\phi} x_i + RT \sum_{i=\mathrm{Ag,Me}} x_i \ln x_i + \Delta^E G_m^{\phi}, \quad (1)$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of the pure component i in the respective reference state with ϕ phase, which is taken from the SGTE pure element database [20]; x_i denotes the mole fraction of the component i; R is the gas constant; T is the absolute temperature; and the term $\Delta^E G_m^{\phi}$ presents the excess Gibbs free energy, which is expressed in the Redlich–Kister polynomial as:

$$\Delta^E G_m^{\phi} = x_{\mathrm{Ag}} x_{\mathrm{Me}} \sum_{m=0}^{n} {}^mL_{\mathrm{Ag,Me}}^{\phi} (x_{\mathrm{Ag}} - x_{\mathrm{Me}})^m, \quad (2)$$

$${}^mL_{\mathrm{Ag,Me}}^{\phi} = a + bT, \quad (3)$$

where ${}^mL_{\mathrm{Ag,Me}}^{\phi}$ is the binary interaction parameter, and the coefficients of a and b are evaluated on the basis of available experimental data.

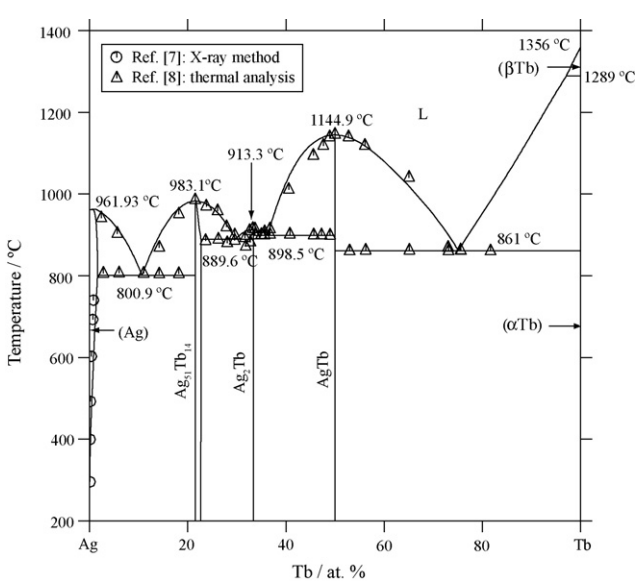

Fig. 1. Calculated phase diagram of the Ag–Tb system compared with the experimental data [7,8].

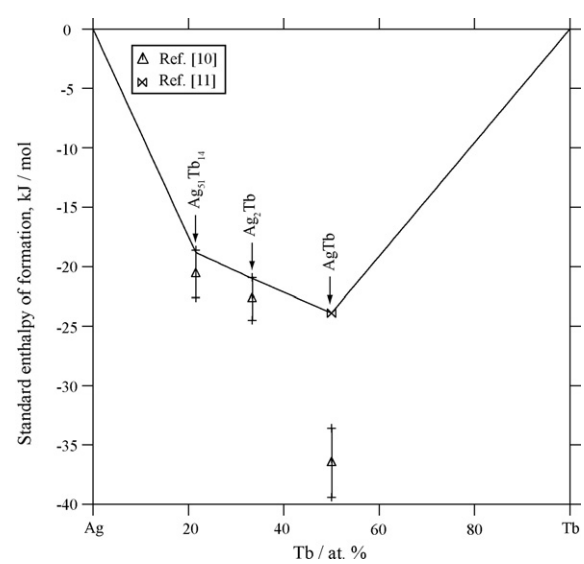

Fig. 2. Calculated standard enthalpies of formation with reference states of Ag (fcc) and αTb (hcp) in the Ag–Tb system compared with the experimental data [10,11].

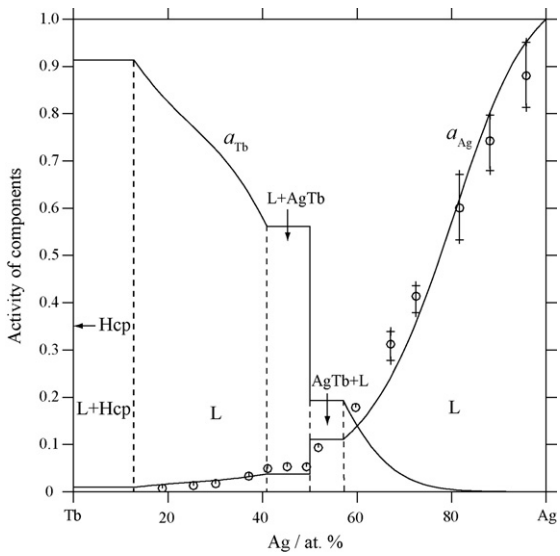

Fig. 3. Calculated activities of components at 1090 °C in liquid Ag–Tb system with reference states of pure elements of Ag (liquid) and Tb (liquid) compared with the experimental data [11].

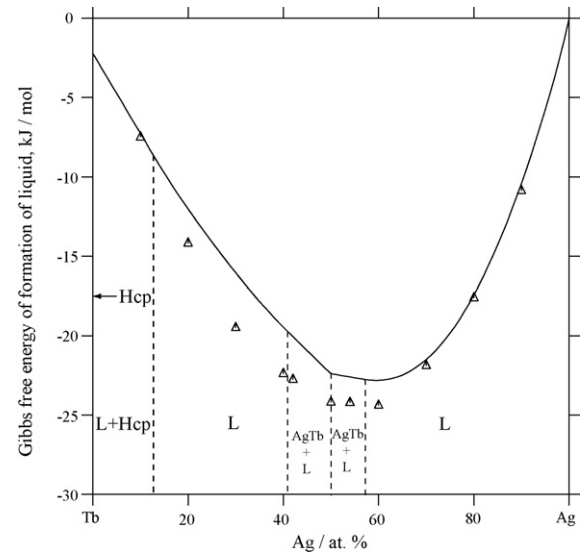

Fig. 4. Calculated Gibbs free energies of formation of liquid at 1090 °C with reference states of pure elements of Ag (liquid) and Tb (liquid) compared with the experimental data [11].

3.2. Stoichiometric intermetallic compounds

The intermetallic compounds of the Ag_2Tb, AgTb, Ag_2Ho and AgHo phases in the Ag–Me (Me: Tb or Ho) binary system are all treated as stoichiometric phases. The Gibbs free energy per mole of formula unit Ag_pMe_q can be expressed, as follows:

$$\Delta^0 G_f^{Ag_pMe_q} = G_m^{Ag_pMe_q} - p^0G_{Ag}^{SER} - q^0G_{Me}^{SER} = a' + b'T, \quad (4)$$

Table 2
Thermodynamic parameters in the Ag–Tb system assessed in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Ag, Tb)
${}^0L_{Ag,Tb}^{Liq} = -97,830 + 30.509\,T$
${}^1L_{Ag,Tb}^{Liq} = -32,453 - 2.468\,T$
${}^2L_{Ag,Tb}^{Liq} = -1223 + 0.125\,T$
${}^3L_{Ag,Tb}^{Liq} = 29,963 - 6.38\,T$
fcc phase, format (Ag, Tb)
${}^0G_{Tb}^{fcc} = {}^0G_{Tb}^{hcp} + 5000$
${}^0L_{Ag,Tb}^{fcc} = 367,756 - 31.663\,T$
${}^1L_{Ag,Tb}^{fcc} = -393,972 - 31.327\,T$
${}^2L_{Ag,Tb}^{fcc} = 15.294\,T$
bcc phase, format (Ag, Tb)
${}^0L_{Ag,Tb}^{bcc} = 50,000$
hcp phase, format (Ag, Tb)
${}^0L_{Ag,Tb}^{hcp} = 50,000$
AgTb phase, format $(Ag)_{0.5}(Tb)_{0.5}$
$\Delta^0G_f^{AgTb} = -23,900 + 1.485\,T$
Ag_2Tb phase, format $(Ag)_{0.667}(Tb)_{0.333}$
$\Delta^0G_f^{Ag_2Tb} = -21,000 + 0.08\,T$
$Ag_{51}Tb_{14}$ phase, format $(Ag, Tb)_{0.785}(Tb)_{0.215}$
${}^0G_{Tb:Tb}^{Ag_{51}Tb_{14}} = {}^0G_{Tb}^{hcp} + 5000$
$\Delta^0G_f^{Ag_{51}Tb_{14}} = -18,800 + 0.846\,T$
${}^0L_{Ag,Tb:Tb}^{Ag_{51}Tb_{14}} = -38,418 + 16.434T + 0.994T\ln T$

where the $\Delta^0G_f^{Ag_pMe_q}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements, and the parameters of a' and b' are evaluated in the present work.

3.3. Intermetallic compound with solubility

The intermetallic compounds of the $Ag_{51}Tb_{14}$ and $Ag_{51}Ho_{14}$ phases with certain solubility have the hexagonal $Ag_{51}Gd_{14}$-type structure. In the present assessment, the Gibbs free energies of the two compounds are modeled by the two-sublattice model $(Ag, Me)_{0.785}:(Me)_{0.215}$. The molar Gibbs free energy of the compounds is presented by the following equation:

$$G_m^{Ag_{51}Me_{14}} = y_{Ag}^I\,{}^0G_{Ag:Me}^{Ag_{51}Me_{14}} + y_{Me}^I\,{}^0G_{Me:Me}^{Ag_{51}Me_{14}} + 0.785\,RT(y_{Me}^I \ln y_{Me}^I + y_{Ag}^I \ln y_{Ag}^I) + 0.785 y_{Ag}^I y_{Me}^I L_{Ag,Me:Me} \quad (5)$$

where y_i^I is the site fraction of element i in the first sublattice. The ${}^0G_{i:j}^{Ag_{51}Me_{14}}$ represents the Gibbs free energies of the $Ag_{51}Me_{14}$

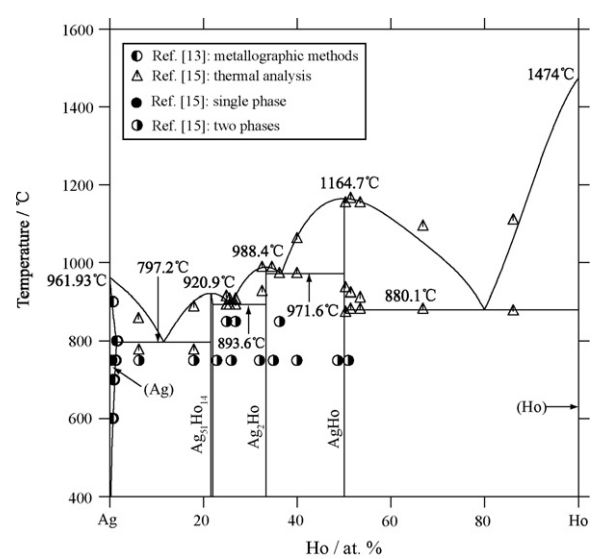

Fig. 5. Calculated phase diagram of the Ag–Ho system compared with the experimental data [13,15].

Table 3
Experimental and calculated invariant reactions in the Ag–Tb system.

Invariant reaction	Reaction type	Composition (at.% Tb)			Temperature (°C)	Reference
L ↔ αTb + AgTb	Eutectic	75.0 ± 1	100	50.0	860 ± 5	[8]
		75.0	100	50.0	860	[9]
		75.1	100	50.0	861	This work
L ↔ AgTb	Congruent		50.0		1145 ± 10	[8]
			50.0		1145	[9]
			50.0		1144.9	This work
L ↔ AgTb + Ag_2Tb	Eutectic	35.0 ± 0.5	50.0	33.3	900 ± 5	[8]
		35.0	50.0	33.3	900	[9]
		36.3	50.0	33.3	898.5	This work
L ↔ Ag_2Tb	Congruent		33.3		915 ± 10	[8]
			33.3		915	[9]
			33.3		913.3	This work
L ↔ Ag_2Tb + $Ag_{51}Tb_{14}$	Eutectic	31.5 ± 0.5	33.3	21.5	885 ± 10	[8]
		31.5	33.3	25.5	885	[9]
		29.7	33.3	22.7	889.6	This work
L ↔ $Ag_{51}Tb_{14}$	Congruent		21.5		985 ± 10	[8]
			21.5–25.5		985	[9]
			21.5–22.7		983.1	This work
L ↔ $Ag_{51}Tb_{14}$ + (Ag)	Eutectic	–	21.5	1.1	801 ± 3	[7]
		11.0 ± 0.5	21.5	1.1	805 ± 5	[8]
		11.0	21.5	1.1	801	[9]
		10.7	21.5	1.6	800.9	This work

compound when the first and second sublattices are respectively occupied by elements i and j. $L_{Ag,Me:Me}$ is the interaction energy between Ag and Me atoms in the first sublattice when the second sublattice is occupied by Me, and is expressed as:

$$L^{\phi}_{Ag,Me:Me} = a'' + b''T + c'' \ln T, \quad (6)$$

the parameters of a'', b'' and c'' are evaluated in this work.

4. Optimized results and discussion

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software [21], which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the importance of data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

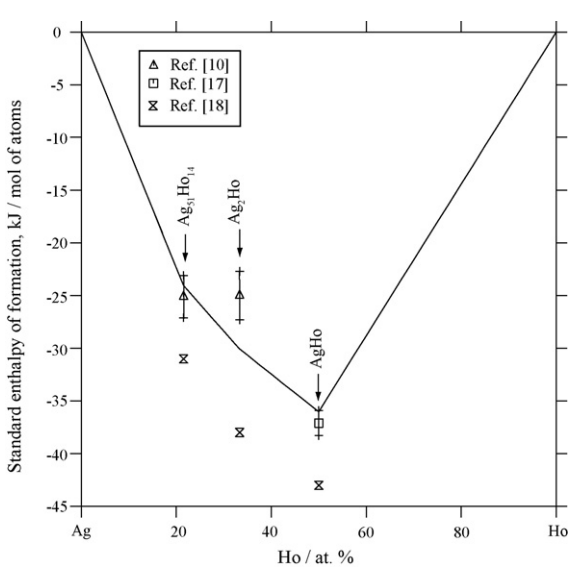

Fig. 6. Calculated standard enthalpies of formation with reference states of Ag (fcc) and Ho (hcp) in the Ag–Ho system compared with the experimental data [10,17,18].

4.1. The Ag–Tb system

The calculated Ag–Tb phase diagram compared with the experimental data is shown in Fig. 1. It is seen that the calculated results are in reasonable agreement with the experimental data [7,8]. The calculated solid solubility of Tb in the fcc (Ag) solution phase is

Table 4
Thermodynamic parameters in the Ag–Ho system assessed in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Ag, Ho)
${}^{0}L^{Liq}_{Ag,Ho} = -117,198 + 17.148\,T$
${}^{1}L^{Liq}_{Ag,Ho} = -50,687 + 1.55\,T$
${}^{2}L^{Liq}_{Ag,Ho} = 23,465 - 23.855\,T$
${}^{3}L^{Liq}_{Ag,Ho} = 102,685 - 47.294\,T$
fcc phase, format (Ag, Ho)
${}^{0}G^{fcc}_{Ho} = 4000 + {}^{0}G^{hcp}_{Ho}$
${}^{0}L^{fcc}_{Ag,Ho} = -59,413 + 3.011\,T$
${}^{1}L^{fcc}_{Ag,Ho} = -22,711 - 0.002\,T$
hcp phase, format (Ag, Ho)
${}^{0}L^{hcp}_{Ag,Ho} = 50,000$
AgHo phase, format $(Ag)_{0.5}(Ho)_{0.5}$
$\Delta^{0}G_{f}^{AgHo} = -36,029 + 3.597\,T$
Ag_2Ho phase, format $(Ag)_{0.667}(Ho)_{0.333}$
$\Delta^{0}G_{f}^{Ag_2Ho} = -30,063 + 0.1276\,T$
$Ag_{51}Ho_{14}$ phase, format $(Ag, Ho)_{0.785}(Ho)_{0.215}$
${}^{0}G^{Ag_{51}Ho_{14}}_{Ho:Ho} = {}^{0}G^{hcp}_{Ho} + 5000$
$\Delta^{0}G_{f}^{Ag_{51}Ho_{14}} = -24,038 + 0.41\,T$
${}^{0}L^{Ag_{51}Ho_{14}}_{Ag,Ho:Ho} = -45,000 + 0.434\,T + 4\,T \ln T$
${}^{1}L^{Ag_{51}Ho_{14}}_{Ag,Ho:Ho} = -21,000$

Table 5
Experimental and calculated invariant reactions in the Ag–Ho system.

Invariant reaction	Reaction type	Composition (at.% Ho)			Temperature (°C)	Reference
L ↔ (Ho) + AgHo	Eutectic	81.0	100	50.0	880	[15]
		81.0	100	50.0	880	[16]
		79.9	100	50.0	880.1	This work
L ↔ AgHo	Congruent		50.0		1165	[15]
			50.0		1165	[16]
			50.0		1164.7	This work
L ↔ AgHo + Ag_2Ho	Eutectic	36.5	50.0	33.3	970	[15]
		36.5	50.0	33.3	970	[16]
		36.6	50.0	33.3	971.6	This work
L ↔ Ag_2Ho	Congruent		33.3		990	[15]
			33.3		990	[16]
			33.3		988.4	This work
L ↔ $Ag_{51}Ho_{14}$ + Ag_2Ho	Peritectic	26.5	22.2	33.3	890	[15]
		26.5	24.4	33.3	890	[16]
		26.0	22.0	33.3	893.6	This work
L ↔ $Ag_{51}Ho_{14}$	Congruent		22.2		920	[15]
			21.5		920	[16]
			21.7		920.9	This work
L ↔ $Ag_{51}Ho_{14}$ + (Ag)	Eutectic	–	22.2	1.6	787	[13]
		–	22.2	1.6	765	[14]
		11.5	22.2	1.6	775	[15]
		11.5	21.5	1.6	795	[16]
		11.5	21.5	1.4	797.2	This work

2.0 at.% Tb, which is close to the experimental data (1.12 at.% Tb) [7]. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Ag–Tb system are given in Table 2, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 3.

The calculated standard enthalpies of formation with reference states of the Ag (fcc) and αTb (hcp) phases compared with the experimental data are presented in Fig. 2. It is seen that the calculated standard enthalpy of formation of the AgTb phase is very close to the value from Ref. [11], but slightly more positive than the value from Ref. [10]. The reason of this difference may be that the element of Tb is very active, and there are some inevitably errors in the process of measurement of standard enthalpy of formation of the AgTb phase [10]. Therefore, the present calculated results are considered to be reasonable. The calculated activities in the liquid binary alloys at 1090 °C are illustrated in Fig. 3 with the experimental data. Fig. 4 shows the calculated Gibbs free energies of formation of liquid phase at 1090 °C. It is seen that an agreement is obtained between the calculated results and the experimental data.

4.2. The Ag–Ho system

The calculated Ag–Ho phase diagram compared with the experimental data is shown in Fig. 5. It is seen that the calculated results are in reasonable agreement with the experimental data [13,15]. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Ag–Ho system are given in Table 4, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 5.

The calculated standard enthalpies of formation with reference states of the Ag (fcc) and Ho (hcp) phases are presented in Fig. 6 with the experimental data. It is seen that the calculated enthalpy of formation of the Ag_2Ho phase is slightly more positive than the value predicted by Niessen et al. [18]. A reasonable agreement is obtained between the calculated results and the experimental data. Fig. 7 shows the calculated enthalpies of mixing of the liquid phase in the Ag–X binary systems. The calculated minimum values in the Ag–Tb and Ag–Ho systems are respectively −25.08 kJ/mol at 43 at.% Tb and −30.17 kJ/mol at 42.5 at.% Ho, which are close to the value (−29 kJ/mol) predicated by Boer and Perrifor [12]. It is seen that the minimum values of enthalpies of mixing of the liquid phase appear at about 40 at.% X in the Ag–X (X: Sc [22], Y [23], Ce [24], Nd [25], Gd [25] and Yb [26]) binary systems.

4.3. The Ag–Tb–Ho system extended from binary systems

The phase equilibria in the Ag–Tb–Ho ternary system are useful for the development of new materials. But no information on the phase equilibria and thermodynamic properties in the Ag–Tb–Ho ternary system is available. In this work, three couples of intermetallic compounds ($Ag_{51}Tb_{14}$-$Ag_{51}Ho_{14}$, Ag_2Tb-Ag_2Ho and AgTb-AgHo phases) are described as solid solutions assuming ideal

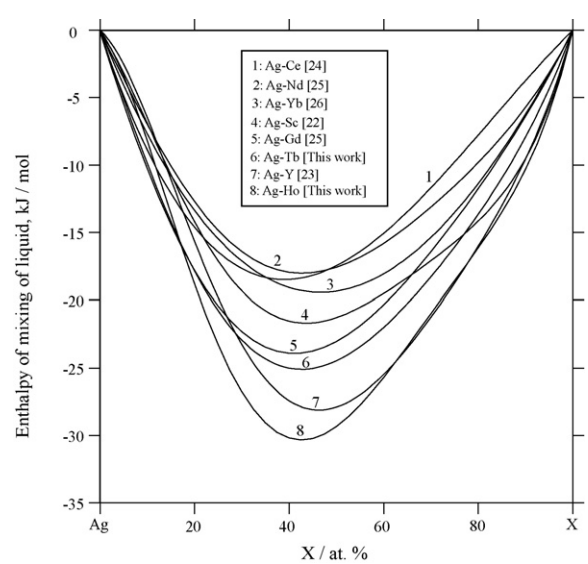

Fig. 7. Calculated enthalpies of mixing of liquid phase with reference states of pure elements of Ag (liquid) and X (liquid) [22–26].

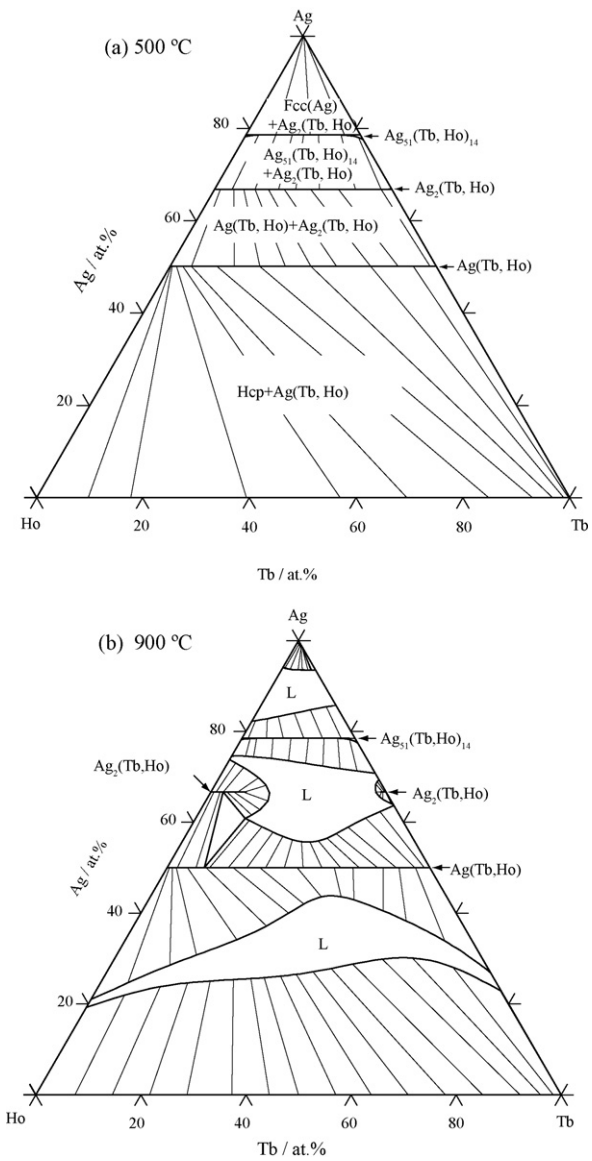

Fig. 8. Calculated isothermal section diagrams of the Ag–Tb–Ho system at (a) 500 °C and (b) 900 °C.

mixing of Tb and Ho. The Ag_2(Tb, Ho) and Ag(Tb, Ho) phases are expressed as Ag_p(Tb, Ho)$_q$, and the solution between $Ag_{51}Tb_{14}$ and $Ag_{51}Ho_{14}$ is described by (Ag, Tb, Ho)$_{51}$(Tb, Ho)$_{14}$. The isothermal section diagrams of the Ag–Tb–Ho system at 500 °C and 900 °C were calculated based on the binary thermodynamic parameters of the Ag–Tb, Ag–Ho and Tb–Ho [27] systems. The calculated phase diagrams in the Ag–Tb–Ho system are shown in Fig. 8. Although the phase equilibria in this system need to be experimentally determined in the future, these predicted phase diagrams give useful information for determination of the phase equilibria by experiments.

5. Conclusions

The phase diagrams in the Ag–Tb and Ag–Ho binary systems were thermodynamically assessed by considering the experimental data including phase equilibria and thermodynamic data. A consistent set of thermodynamic parameters has been optimized and most of the experimental information can be satisfactorily reproduced on the basis of the optimized thermodynamic parameters. In addition, the phase diagrams of the Ag–Tb–Ho ternary system were predicted for guiding the experiments in the future.

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